Original Article

Comparison of the Accuracy of Three Instruments (APACHEII, SOFA, and MPMII) in Predicting the Mortality of Patients in Intensive Care Units

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ABSTRACT

Accurate prediction of mortality in intensive care units (ICUs) is vital for patient management and resource allocation. Among the most widely used scoring systems are APACHE II, SOFA, and MPM II, each providing valuable but variable estimates of patient outcomes. The present study aimed to compare the accuracy of these three instruments in predicting mortality in ICU patients. This cross-sectional study (2020) included 150 ICU patients from Ardabil University hospitals. Eligible participants were adults (>18 years) with ICU stays ≥24 hours, excluding burn and cardiac surgery cases. Data were gathered using demographic forms and APACHE II, SOFA, and MPM II questionnaires. Statistical analyses (SPSS v23) applied descriptive statistics, t-test, correlation, sensitivity, and specificity. Among the studied patients, 85 were male, with a mean age of 60.74 ± 19.62 years. The average duration of hospitalization was 10.17 ± 10.34 days, and 52 patients died. Logistic regression showed that hospital type, gender, age, length of hospitalization, duration of intubation, diagnosis, and GCS were not significant predictors of mortality. Predicted mortality was 32.21% with APACHE II, 32.50% with SOFA, and 64.01% with MPM II. Sensitivity and specificity values were 83% and 48% for APACHE II, 79% and 46% for SOFA, and 73% and 50% for MPM II, respectively. In conclusion, APACHE II demonstrated higher accuracy compared with SOFA and MPM II, and its application is recommended as the preferred prognostic tool in ICU settings.

Introduction

Care is defined as the primary duty of nursing, and it should be considered at three different levels of health. One of the important departments providing advanced and specialized care is the special care unit in every hospital. The special care department is a specialized department in every hospital where patients with high care needs and special medical services are admitted. If timely services are not provided for these patients, many problems are added to the patient's issues [1]. The most important goals of the special department are to preserve the patient's life and provide high-quality care for the patient. The quality of care in the special department can have a significant impact on the

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death rate of patients. About 13% of hospital expenses and 2.4% of health expenses are related to care provided in the ICU [2]. According to global estimates, the death rate in special departments is between 6.4% and 40 % [1]. More than two million deaths occur annually in the United States, which can be reduced by more than 30% with basic care in the intensive care unit. The improvement of treatment capacities and modern technologies in recent years has increased the demand for the use of special care centers. Therefore, predicting mortality in special care units is very important in facilitating the management of hospital resources and evaluating the quality of care [3]. Considering the high cost of special departments and limited facilities, it is necessary to be careful to make better use of the facilities and their effectiveness. For this reason, classification of patients and accurate prediction of the discharge times can prevent unnecessary admissions in the special care department, which leads to rational and appropriate use of intensive care units [4]. By optimizing the use of the facilities in special departments, it is possible to provide services to more needy patients, and it is possible to categorize the services to improve the quality. For this purpose, knowing the needy patients and predicting the service outcomes can be very helpful [4]. Evaluation of the results of medical treatments began in 1863, when Florence Nightingale first reached to this conclusion [5].

Thus, in the past thirty years, various tools have been created to classify patients in terms of disease severity and mortality. These tools serve as a guide for the nursing and medical teams to assess the patient's condition. The purpose of all these tools is to accurately evaluate and continue the process of treating the patient by creating the best care and clinical conditions. Using new assessment methods, it is possible to determine the need for care and the severity of the disease in a more accurate way than before this, distinguishing patients who require high care from other patients to provide better and more precise services that reduce their mortality rate. In 1981, the first classification system named APACHE was proposed by the George Washington University Medical Center as a method to measure the severity of the disease. In 1985, the initial version was revised and a simplified version named APACHE II was created by Naus *et al.* This version includes twelve acute and chronic physiological variables that are filled in the special section of the relevant form within the first 24 hours of admission. Its score ranges from 0 to 71; a higher score indicates the deterioration of the patient's condition [2,6]. APACHE II is widely used in special care centers today. This tool requires high variables, and today this problem has been resolved by software calculation [7]. One of the advantages of this tool is its simplicity in its use and the small number of variables compared to other versions [8]. This tool is more useful in ranking the risk of mortality in clinical research and making decisions in critical situations [2,6]. After years, the third and fourth versions of Apache were also developed. However, due to having many variables and the lack of certain tests required by the tools in the studied research centers, they were not used in this study. The SOFA scoring system was created in 1994 and updated in 1996, which evaluates six organs: pulmonary, blood, cardiovascular, liver, central nervous system, and kidney. Each system is score from 1 to 4, and the score ranges from 0 to 24. The highest score of SOFA indicates a high mortality rate. The SOFA form is filled out and calculated after the first 24 hours of admission, and then the relationship between their score and the patients' output is studied [8]. One of the advantages of the SOFA tool is that it does not need to be calculated by computer software, the simplicity of the calculation, and its low cost [9]. A few years ago, it was suggested that old models should not be used for a long time because the inaccuracy of their results can increase. In 1985, another tool named MPM II was invented, in which the mortality rate of patients is estimated using physiological disorders.

In 1993, this control tool and the MPMII model were invented. This is the most common model of this tool. This model includes before admission and 24 hours after admission. The MPM II model includes 15 variables and the 24 hours post-admission model includes 13 variables. Apart from the age variable, all variables are answered as yes or no. One of the benefits of this model is its ability to be compared with tools such as APACHE. Another advantage of this tool is the ease of use, which

reduces the time spent for examining the patient, eliminates the need for laboratory data, and predicts the length of the patient's stay in the special department. In developed countries, many research studies have been conducted on these scoring systems, and have increased their practical use. In Iran, less research has been done on this matter. Comparative studies of the performance of these tools are also very limited in Iran. With regard to cases such as the existence of diversity in tools for rating patients in terms of the type of care provided, the use of old methods or GCS, as well as the application of SOFA and APACHE scoring in the new evaluation sheets of specialty departments. Also, a lack of research in this field and the limited comparative studies of the effectiveness and performance of these tools in Iran suggest the need for conducting the research on the comparison of these tools and to achieve the level of accuracy of each of them. The purpose of this study is to compare the three tools APACHE II, SOFA, and MPM II in the special care units of Ardabil Medical Education and Training Center in 2016 to achieve their accuracy compared to each other was done.

Methods

The present study is a descriptive-analytical study that was conducted on 150 patients hospitalized in the special departments of neurology, neurosurgery and internal surgery of Fatemi and Alavi hospitals in Ardabil City during 2020. The sample size for this study was 150 patients which calculated based on Cochrane formula with a 95% confidence interval, d=0.05 and power 80% using the formula $Z^2*p*(1-p)/(d^2)$. The sampling method was simple random sampling and all cases selected from all referred patients to the hospitals. All three tools along with demographic characteristics were completed by the researcher and two trained and coordinated evaluators after coordination with the head of the department and request for coordination to conduct the necessary research tests in addition to routine cases and according to the entry conditions. The necessary training for the evaluators was provided during a one-day session.

All eligible patients admitted to the ICU were included in the study. Demographic data such as age, sex, medical history, and current diagnosis were completed by recording from the patients' files. The Apache tool questionnaire, which includes three main parts. The first part determines to determine the Glasgow Coma Scale in patients, the second part assesses the risk of death in patients, and the third part evaluates to determine the acute physiological state of patients. This questionnaire was filled in twenty-four hours after admission to the special department and only once. In this research, the MPM II tool was used. The SOFA tool, which includes the assessment of six vital organs, was completed in the first twenty-four hours after admission. All three tools were filled for each patient. The follow-up period was from entering the special ward until leaving the ward. The way of follow-up method was such that if the studied patient died in the special ward, he was included in the category of deceased. However, if the patient was discharged from the ward alive, one month later the condition of the patient would be followed up by phone with the patient himself or his family, who were considered alive, except for the group of those saved from death. After calculating the scores, the relationship between the actual mortality and the predicted death was examined. This research was registered and implemented by the ethics committee of Ardabil University of Medical Sciences with the ethical code IR.ARUMS.REC.1396.227.The consent forms were completed for all patients.

The written consent form was filled by the patient himself or his legal guardian. The information obtained from the checklists was entered into SPSS v23 and data analysis and significant correlations between the data were performed using Chi-independent t-tests, logistic regression to detect effective factors on mortality and Pearson's correlation. The results of the data analysis were expressed in the form of tables and graphs. In this study, the ROC curve was not used, but w the sensitivity and specificity of all three indices were evaluated based on necessary information from models and confusion matrix. In this analysis, the significance level was considered less than 0.05.

SOFA Questionnaire

This evaluation tool includes six vital organs: lungs, blood, cardiovascular, liver, central nervous system, and kidneys, and each organ is given a score from 0 to 4. The sum of these numbers from the six vital organs is the patient's daily score. The range of questions is divided into five options and from normal to abnormal; normal receives a score of 0, and abnormal receives a score of 4. The total scores of the SOFA scoring system range between 0 and 24. Patient mortality increases when their scores are high.

MPM II Questionnaire

In the first part of the tool for predicting death, seven choices were examined in the form of two options (yes and no), including admission from the emergency department, cardiopulmonary resuscitation before admission, cancer, chronic kidney disease, infections, hospitalization history. The patient was in the intensive care unit for the last six months and was hospitalized in the intensive care unit after surgery. In the second unit, factors such as systolic blood pressure, heart rate, level of consciousness of the patient, and the patient's age were checked and recorded.

APACHE Questionnaire

APACHE II consists of three parts: the first part includes 11 items to measure physiological variables and Glasgow Coma Scale. The minimum score for each question in the physiological dimension is zero and the maximum score is four.
The second part is the score that the patient receives based on their age group in this tool. For patients under the age of 45, the minimum score in this part is 0 and the maximum is 6. The third part is related to the evaluation of chronic diseases or chronic and severe failures of one or more organs in the patient, including the heart, lungs, kidneys, liver and immune system. The total points obtained from these three parts are added together and the APACHE II score is obtained, with the minimum score being 0 and the maximum being 71. Based on this score, the mortality rate of patients is predicted as a percentage.

Results

The average age of the patients was 60.74 ± 19.62 years with an age range of 19 to 93 years. The majority of patients with 79 cases (52.7%) were in the age range of more than 65 years and the least number of patients were in the age range of less than 20 years. Additionally, the majority of patients (56.7%) were men. The average duration of intubation for patients it was 6.97 ± 9.01 days, and the duration of hospitalization for the patients was 10.34 ± 10.17 days.
32 patients (21.2%) had breathing problems (Figure 1).
The highest number of deaths among patients aged 65 years and older was 29 cases (55.8 percent). The total number of deaths of patients was 28 cases (32.9 percent) in men and 24 cases (36.9 percent) in women. The observed mortality in patients was 52 cases (34.7 percent). There was no significant relationship between the age, gender, and life status of patients and the mortality rate of patients. The average percentage of mortality predicted using the MPM II questionnaire at baseline was equal to 22.86 ± 21.98 and in the following 24 hours it was equal to 50.51 ± 39.31 in patients. The mean difference between the two groups was significant (p=0.023) (Table 1).
A significant correlation was found between the observed mortality rate and the predicted mortality rate by the MPM II questionnaire at zero hour (r = 0.208, p = 0.011) and at 24 hours (r = 0.251, p = 0.002). The arterial oxygen pressure of patients was less than or equal to 200 in 65 cases (43.3 percent). The platelet level of patients was in 96 cases. (0.64 percent), which was more than 150 thousand. The bilirubin level of patients was reported to be less than 1.2 in 128 cases (85.3 percent). According to the SOFA questionnaire, in 79 cases (52.7%) of the patients, the level of creatinine was reported to be less than 1.2. Based on the SOFA questionnaire, in 45 cases (30.0%) of the patients, the level of consciousness was reported between 6 and 9. Additionally, 125 cases (83.3%) of the patients did not have hypotension.

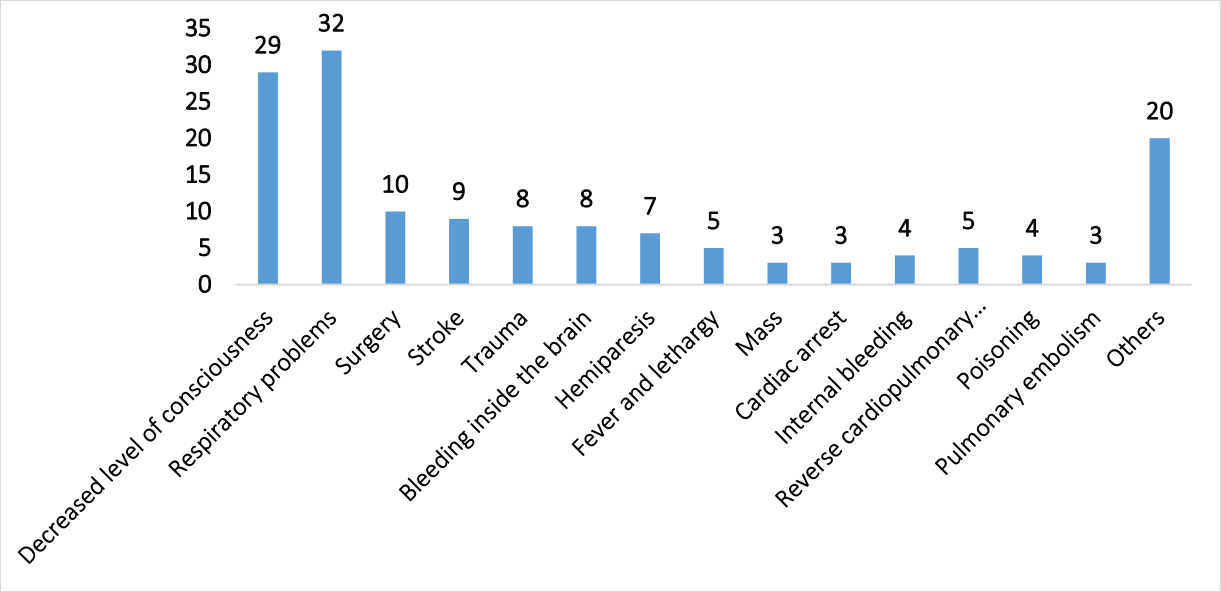


Figure 1 Frequency of clinical characteristics of the studied patients based on initial diagnosis

Table 1 The mortality predicting rate in baseline and 24 hours late based on MPMII

Death risk	MPM II at baseline		MPM II 24 hours late	
	n	%	n	%
<20	93	62	54	36
20-40	28	18.7	21	14
41-60	13	8.7	10	6.7
61-80	13	8.7	9	6
>80	3	2	56	37.3
Mean	22.86 ± 21.98		50.51 ± 39.31*	
Total number	150	100	150	100

Independent two sample t-test; *p<0.05

The highest risk of death in the risk group was 22% with 52 cases (34.7%). The average percentage of death risk predicted by SOFA questionnaire was 26.51 ± 22.53. A significant relationship between the death rate and the observed mortality was obtained with the predicted mortality rate by SOFA questionnaire ($r = 0.254$, $p = 0.001$). The highest observed mortality rate was 52 cases (34.7%) in the group with a percentage of death risk of 22%. The results showed that with the increase in SOFA score up to 7, the percentage of mortality also increases, and then decreases (Table 2).

The mean of predictive death rate in deceased patients with 64.01 in MPMII was higher than other two methods. Additionally, the differences in mean score of three indices between both live and deceased patients were statistically significant (Table 3).

The GCS score of the patients was examined as one of the items in the APACHE II questionnaire. The highest frequency with 89 cases (59.3%) was in the range of 3-8. A significant correlation was found between the observed mortality rate and the predicted mortality rate by APACHE II questionnaire ($P = 0.007$) ($r = 0.235$, $p = 0.007$).

The Pearson correlation comparison of three tools shows that the correlation between the predicted mortality rate and the observed mortality rate in the SOFA tool is stronger than the other two tools, and the MPM II tool is stronger than APACHE II.

The logistic regression model was used to investigate the factors affecting the mortality rate and it was found that none of the factors-hospital type, gender, age, and length of

hospitalization, duration of intubation, diagnosis, and GCS had any effect on mortality. In the examination of the accuracy of the three methods, it was found that the detection power of MPM with an average of 64% was higher than the SOFA and APACHE II methods, and it had high accuracy. However, the two methods SOFA and APACHE II were almost the same in terms of effectiveness and accuracy with an averages of 32.50% and 32.21%, respectively (Table 4).

Table 2 The rate of death risk in patients by SOFA method

Final score	Death risk (%)	n	%
0-1	0	9	6
2-3	7	19	12.7
4-5	20	28	18.7
6-7	22	52	34.7
8-9	33	24	16
10-11	50	11	7.3
>11	95	7	4.7
Mean ± SD		26.51 ± 22.53	
Total		150	100

Table 3 Mean of predictive death rate based on three indices by live status of patients

Live status	Live			Dead			p-value*
Predictive death rate	n	Mean	SD	n	Mean	SD	
MPMII in 24 hour	98	43.35	38.97	52	64.01	36.63	0.002
SOFA	98	21.98	15.39	52	32.5	23.79	0.005
APACHEII	98	23.95	16.76	52	32.21	18.7	0.009

Independent two sample t-test ; *p<0.01

Table 4 The rate of death risk in patients by APACHE II method

Final score	Death risk (%)	n	%
0-4	4	6	4
5-9	8	21	14
10-14	15	29	19.3
15-19	25	52	34.7
20-24	40	27	18
25-29	55	8	5.3
30-34	75	2	1.3
>34	85	5	3.3
Mean ± SD		23.98 ± 21.43	
Total		150	100

The relationship between the three indicators was determined using Pearson's correlation coefficient, that the relationship between SOFA and APACHE II indices with MPM was positive and significant, such that between MPM and SOFA r=0.23 and P = 0.005, MPM and APACHE II r = 0.35 and P = 0.001 and between SOFA and APACHE II r = 0.44 and P = 0.001 (Figures 2-4). In the analysis of the sensitivity of the above three methods using the standard table, it was determined that MPM II has a sensitivity of 73% and a specificity of 50%, SOFA has a sensitivity of 79% and a specificity of 46%, and APACHE II has a sensitivity of 83% and a specificity of 48%. APACHE II has greater sensitivity in detecting mortality than the other two methods. In the examination of the characteristics of MPM II, it has the most characteristics compared to the other two methods. The sensitivity of APACHE II with 83% was higher than other two indices. (Table 5).

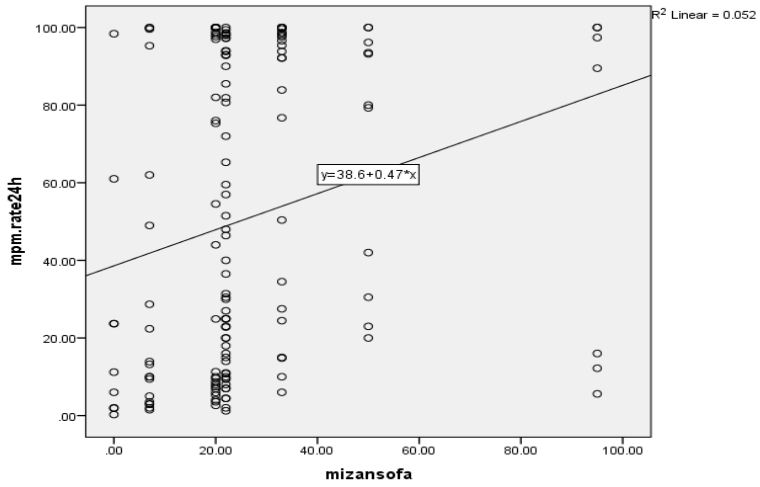


Figure 2 Correlation chart of SOFA and MPM II

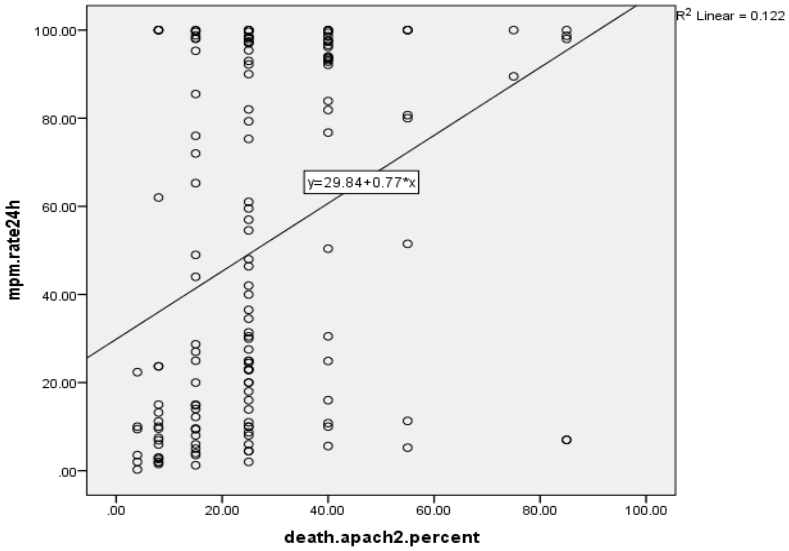


Figure 3 APACHE II and MPM II correlation chart

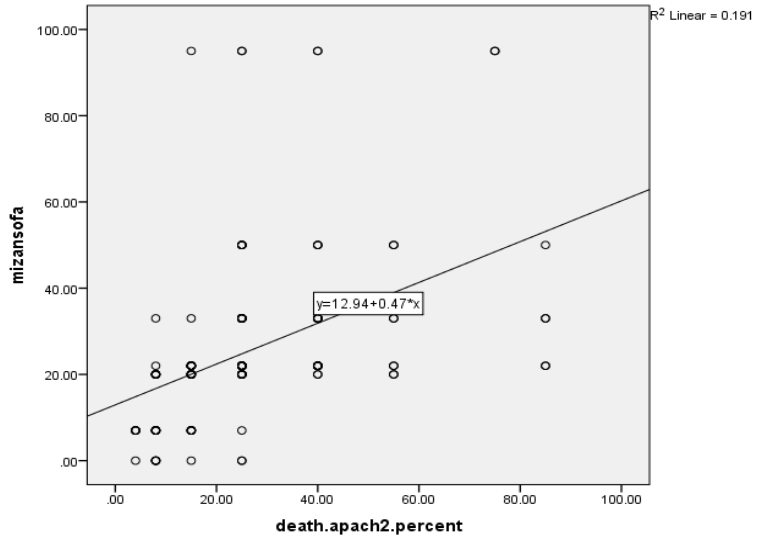


Figure 4 APACHE II and SOFA correlation diagram

Table 5 The sensitivity and specificity of three indices

Method Accuracy	APACHEII (%)	SOFA (%)	MPMII (%)
Sensitivity	83	79	73
Specificity	48	46	50

Discussion

The aim of this study was to compare the accuracy of three tools APACHE II, SOFA and MPM II in predicting the mortality of patients hospitalized in the special care units of medical education centers in Ardabil City. In this study, 150 patients who met the inclusion criteria were included in the study. The average age of these individuals was 60.74 ± 19.62 years. Additionally, the gender of the patients was checked and it was found that 65 patients (42.2%) were female and 85 patients (55.2%) were male. In the study of Soleimani *et al.*, which was conducted on 240 patients of special wards in Qazvin City in 2009, the average age of the patients was 48.78 ± 19.90; 171 patients (71.3%) were male and 69 patients (28.7%) were female [4]. In the study of Yaqoubi *et al.*, Qazvin, among the patients included, the average age reported was 44.25 ± 23.09 years (with an age range between 2 and 86 years), and 52 patients (59.1%) were male while 36 patients (40.9%) were female [10].

The duration of hospitalization of the patients was calculated and the average of this duration was 10.34 ± 10.17 days. The duration of intubation was calculated and the average of this duration was 6.97 ± 9.01 days. In the study by Soleimani *et al.*, which was conducted on 204 patients in the intensive care unit in Qazvin City, the average duration of hospitalization in the intensive care unit was 8.94 ± 13.84 days and 33.3% of patients needed ventilator and special respiratory care [6]. In the study by Mousavi *et al.*, which was conducted on 110 patients in Qom City in 2011, 66% of patients were reported to be hospitalized for less than 5 days, while in this study, 48.7% of patients were within 5 to 10 days. The total mortality of patients observed in this study was 52 cases (34.7%). In this study, the frequency of patient deaths by gender was also investigated, and the highest rate of patient deaths was in men with 28 cases (53.8 percent). In this study, the frequency of patient deaths by age groups was investigated that the most deaths of patients in the age range of more than 65 years with 29 cases (55.8%) are included. In the study by Ala *et al.*, which was conducted in 2014 at Hazrat Rasool Akram Hospital in Tehran, the death rate of 90 patients (30%) was reported out of a total of 300 patients [11]. In the study by Karami *et al.*, the reported mortality rate was 81 people (43.7 percent) [12]. In Husseini *et al.*'s study, the death rate was 82 patients (27.3%) and the average age of the deceased was 66.41 ± 22.50, compared to the average age of the survivors was 49.59 ± 25.60. This significant difference was reported [5]. However, in the present study, there was no significant difference between patient age and mortality. In the study of Schoe *et al.*, which was conducted in 2014 on 269 patients hospitalized in the special care department, the mortality rate of 73 patients (27%) was reported, and the average age of the deceased patients was 8. There was a significant difference between the average age of deceased and living individuals ($P < 0.001$), which was not consistent with our study [13]. In the study by El-Naggar *et al.*, which was conducted on 110 patients admitted to Medina City Hospital, the observed mortality rate was 27.3% [14].

In the study by Aminiahidashti *et al.*, involving 88 patients who were hospitalized in intensive care units, the average age of the patients was 53.45 ± 20.37 and 65.9% of them were men. The average stay of patients in the care department was 4.77 ± 3.70 days and 48% of patients died. In this study, similar to the present study, there was no significant difference between the deceased individuals and the gender of the patients ($P = 0.53$) [15]. In the present study, there was no significant difference between the mortality rate and age or length of hospitalization. In the study by Soleimani *et al.* which was conducted on 240 patients in Qazvin City, the data analysis showed that there was no significant difference between the patients' ages and the mortality rate, which is algin with the

results of the present study [4]. Sasikala *et al.* conducted a study in Turkey on stroke patients, which lasted from January 2008 to February 2009. Fifty-five patients on ventilators were included in the study. 52.6% of patients were male and their average age was 76.5 ± 11.5 years. The average age of the women studied was 72 ± 5. Actual mortality was reported as 34.54%. The sensitivity and specificity of APACHE IV were 94.7% and 94.4%, respectively, while APACHE II recorded 100% and 86.1%, respectively. Similar to this study, there was no significant difference between gender and the mortality rate of patients (P = 0.146) [16]. The predicted risk of death of the patients was investigated based on the APACHE II questionnaire. The highest frequency of the risk of death was 25% in 52 cases (34.7%) of the patients. The highest observed mortality rate was in 19 cases (36.5%) in the group, with a death risk percentage of 22%. The average percentage of death risk predicted by the questionnaire was 23.98 ± 20.43. Additionally, in this study, a significant correlation was found between the observed mortality rate and the predicted mortality rate by APACHE II questionnaire (P = 0.004). Also, the Pearson correlation was reported as 0.235. The sensitivity and specificity of APACHE detection in this study were measured as 83% and 48%, respectively. In Yaqoubi *et al.*'s study, the average mortality risk percentage obtained by the APACHE II questionnaire was 19.88 ± 15.1 [10].

In Mohammadi *et al.*'s study, the average score of deceased patients was 24.5 ± 1.8 and the average score of living patients was 21.8 ± 1.7 [17]. In the study of Soleimani *et al.*, the average score of APACHE II in living patients was 13.97 ± 1.51 and in deceased patients was 25.36 ± 0.84. The overall average risk score of the patients was 11.38, the predicted death rate was 30.26% and the observed death rate was 27.9% [6]. In Kim *et al.*'s study, the average score of APACHE II was 11.75 ± 6.9 in living individuals, while the average score in deceased patients was 22.06 ± 7.9, showing a difference was statistically significant (P = 0.001) [18]. In Sasikala's study, the observed mortality rate was 34.54%, with the sensitivity of APACHE II diagnosis at 100% and its specificity at 86.1% [16]. In the study by Aminiahidashti *et al.*, the mean score APACHE II was reported as 16.38 for living patients and 23.17 for deceased patients. In the study by Kim *et al.*, the average score of APACHE II in living patients was reported to be 41.00, while the average score for deceased patients was 0.83, revealing a significant difference was found between the two (P < 0.001); this study also found 26.7% mortality in patients [18]. In the study by El-Naggar *et al.*, the percentage of mortality predicted by the APACHE II scale was reported as 8.7% in living patients and 60.5% in deceased patients. In these two scales, a significant level (P < 0.001) has been obtained [14], which is consistent with the present study. According to SOFA questionnaire, the highest risk of death in patients was 22% with 52 cases (34.7%). The average percentage of death risk predicted by SOFA questionnaire was 26.51 ± 22.53. A significant correlation was found between the observed mortality rate and the predicted mortality rate by the SOFA questionnaire (P = 0.001). Additionally, the Pearson correlation was reported as 0.274. The study by Hosseini *et al.* evaluated the SOFA scoring system in predicting the outcomes of patients hospitalized in visa wards and 82 of the 300 patients studied (27.3%) died. The sensitivity and specificity of SOFA were 57.3% and 67%, respectively. According to this research, a significant correlation was found between the predicted mortality rate and the observed mortality rate (P = 0.001) [5].

The risk of mortality in patients was evaluated using the MPM II questionnaire in the first hour and in the next 24 hours. The average percentage of predicted mortality at zero hour was predicted as 22.86 ± 21.98. Also, the average percentage of predicted mortality in 24 hours in patients was 50.51 ± 39.31; the lowest percentage of predicted mortality was 0.27 and the highest percentage of death was 0/100 was obtained. In the study by Soleimani *et al.* (2011), the average death predicted using the MPM tool was 39.62 ± 39.37% (minimum 1.37 and maximum 99.73) and the average observed death was reported as 33.7% [4]. The relationship between the mortality rate of patients and the percentage of mortality predicted by the MPM II questionnaire was investigated. There was a significant correlation

between the percentage of patient mortality and the predicted mortality at zero hour (P < 0.011) and was obtained at 24 hours (P < 0.002). In the study by Soleimani *et al.* a significant relationship was found between observed mortality and predicted mortality, with a reported level of significance (P = 0.013) [4].

The percentage of mortality predicted by the MPM II questionnaire was investigated concerning the gender of the patients. There was no significant difference between the gender of the patients and the predicted mortality rate (P = 0.812) and (P = 0.073). Additionally, in this study, the correlation between the observed mortality rate and the predicted mortality rate by the MPM II questionnaire in the first hour (P < 0.011) and in the 24th hour (P < 0.002) yielded 0.208 and 0.251, respectively. In Kim *et al.*'s study, a significant difference was found between the Glasgow Coma Score of patients and the observed mortality rate (P = 0.001), as well as a significant difference between the duration of intubation of patients in the first hour and 24 hours, which was then observed with the death rate of patients (P < 0.001) [18]. But in the present study, no significant relationship was found between the level of consciousness and the duration of intubation. In the comparative study of Nagvi *et al.* on the three tools APACHE II, SOFA, and SAPS II, which was conducted in Pakistan, 96 patients with the same gender distribution were included in the study. The average Apache score in deceased individuals was 27.97 ± 8.53 higher than the living ones (15.82 ± 8.79) with statistically significant value (P < 0.001). The average score of SOFA in the deceased (9.86 ± 4.88) was higher than that of the living (5.63 ± 3.63) with a significant level (P < 0.001) [19].

In a comparative study between SOFA, APACHE II and SAPS tools that was conducted in Korea in 2013, 149 organophosphate poisoned patients were included in the study from September 2006 to December 2012. In this study, the sensitivity and specificity of SOFA were reported as 86.2% and 82.4%, respectively, and the sensitivity and specificity of APACHE II were reported as 65.5% and 68.6%, respectively. The result of the study reported SOFA with an accuracy rate of 83.2% ahead of APACHE II, which had an accuracy rate of 79.4% [20]. This research was not consistent with the result of our study. In the study of Czajka *et al.*, which compared the three instruments APACHE II, MPM II, and SAPS in Turkey, the average mortality of patients was determined to be 53.93%. In APACHE II, the predicted death rate was 60.62% and in MPM II it was 46.89% [21]. The limitation of the capacity of the special care department sometimes led to the admission of sick patients in other departments, where the study was not possible for the researcher. Also, it was practically impossible to determine the bed occupancy rate in special departments.

Conclusion

According to the results and the comparison between the SOFA, MPMII and APACHE II questionnaires, it was determined that the APACHE II tool with the highest sensitivity and specificity, has the greatest accuracy in predicting the mortality of special wards patients. The results of the research help caregivers in special care departments to use a tool that is more accurate and efficient. It is suggested that a study be conducted with a larger statistical sample.

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Conflict of Interest

No conflicts of interest were reported by the authors in this study.

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